

Research report

Affective temperaments in clinically-well subjects in Turkey: initial psychometric data on the TEMPS-A

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Abstract

Background: This is a first attempt to evaluate the reliability and factor structure replicability of the Temperament Evaluation of Memphis, Pisa, Paris and San Diego Autoquestionnaire (TEMPS-A) in its Turkish Version. The questionnaire is a self-report 110-item measure that postulates five affective temperaments—the depressive, cyclothymic, irritable, hyperthymic, and anxious—which embody both strengths and liabilities along affective lines. **Methods:** The questionnaire was administered to 658 clinically-well subjects in a Turkish university circle. We undertook item analysis and test–retest reliability. We then examined internal consistency through factor analysis with PCA rotation. **Results:** We found good to excellent test–retest reliability (0.73–0.91), and internal consistency (0.77–0.85). We deleted 10 items with factor loading <0.20 for their own subscales, resulting in a questionnaire with 99 items. Despite considerable overlap between depressive and cognitive anxiety traits, a distinct “nervous”—anxious factor emerged as well, and the hypothesized (original English) 5-factor structure of the TEMPS-A was supported. Cut-offs for each temperament were based on z-scores higher than +2S.D. Dominant irritable (3.7%), nervous–anxious (3.7%) and depressive (3.1%) temperaments were the most common in this population, whereas dominant cyclothymic (1.7%) and hyperthymic (1.2%) temperaments were relatively uncommon. These temperaments tended to lose their intensity with age. As expected, women scored significantly higher on the nervous–anxious, and men on the hyperthymic temperaments. **Limitations:** The sample was composed of younger subjects with higher education than the general population of Turkey. Although the distribution of the scores for each of the temperaments deviated somewhat from normal curves, for heuristic reasons we did attempt to provide prevalence rates based on z-scores. **Conclusion:** In this preliminary version of the TEMPS-A, we have retained 100 (of the original 110) traits loading >0.20. Some deleted items referred to sleep, others appeared socially desirability traits in the Turkish culture endorsed by many subjects. Nonetheless, item analyses within each factor revealed traits indicative of personal assets (specific to each temperament) along with those which might represent vulnerability to affective illness. This is in line with the hypothesized original theoretical framework of the senior authors. Even in this “first pass,” in its Turkish version the TEMPS-A is a

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reliable and valid instrument. Further refinement of the instrument will require the study of a nationally representative sample in Turkey.

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1. Introduction

Although origins of the concept of temperament can be traced back to Hippocratic times, [Kraepelin \(1899\)](#) was the first to formally introduce it into clinical psychiatry. The concept of temperament was popular until the psychodynamic preference for personality disorders overshadowed this classical perspective during the second half of the 20th century. The revival of interest in temperament in contemporary psychiatry is relatively recent ([Akiskal et al., 1979](#)).

This research builds on the work of the last author (H.S.A.)—originally based in Memphis—who operationalized depressive, hyperthymic, irritable, cyclothymic and anxious temperaments as the proximal behavioral phenotypes in the pre-morbid course of affective disorders ([Akiskal, 1989](#); [Akiskal and Akiskal, 1992](#); [Akiskal and Mallaya, 1987](#)). Following [Kraepelin \(1899\)](#); [Kretschmer \(1936\)](#), this body of work hypothesized that these temperaments represent the subclinical foundations of affective disorders, and that affective temperament represents an early clue for a recurrent mood, pre-bipolar disorder ([Akiskal, 1995](#)). As for interpersonal relationships and life events, they have been proposed as mediating mechanisms between temperament and clinical mood episodes ([Akiskal, 1996](#)).

Temperament, character and personality are all overlapping constructs referring to the basic nature of a person ([von Zerssen and Akiskal, 1998](#)). Temperament is believed to be more “heritable,” and relatively stable throughout life. To measure affective traits which could be stable over the lifespan, the senior author (H.S.A.) and his colleagues first studied subthreshold mood and neurotic disorders, leading to the delineation ([Akiskal, 1979](#); [Akiskal, 1983](#)) and operationalization ([Akiskal and Mallaya, 1987](#); [Akiskal, 1998](#); [Akiskal and Akiskal, 1992](#)) of affective temperaments that could be useful in clinical prac-

tice. Subsequently, in collaboration with psychiatrists and primary care physicians in Memphis, Pisa, Paris, and San Diego during the early 1990s, they developed these temperaments into an interview (TEMPS-I) ([von Zerssen and Akiskal, 1998](#); [Placidi, 1998](#)), and eventually into a self-administered Autoquestionnaire or Temperament Evaluation of Memphis, Pisa, Paris and San Diego Autoquestionnaire (TEMPS-A) format consisting of five subscales ([Erfurth et al., 2005](#)). Each of the subscales consist of affect-laden items, as well as socially adaptive traits ([Akiskal and Akiskal, 1988, 1992](#)).

The aim of the present report is to examine the reliability of the Turkish version of the self-report TEMPS-A and to replicate its postulated 5-factor structure. A related objective of ours was to provide preliminary cut-off points of the each of the subscales, and their relative prevalence in the Turkish population under study.

2. Methods

2.1. Measures

The TEMPS-A is a self-report instrument consisting of five subscales. Its 110 constituent items inquire about the subject’s whole life about traits along depressive, cyclothymic, hyperthymic, irritable, and anxious lines.

2.2. Translation

The questionnaire was translated into the Turkish language by two bilingual psychiatrists. The questionnaire was back-translated into English by a third bilingual psychiatrist. The originator of the instrument (H.S.A.) evaluated the back-translated version of the measure, and an identical translation process was carried out for the items that required revisions.

Table 1
The results on reliability and internal consistency

	Test–retest correlations	Cronbach-Alpha coefficients
Depressive temperament	0.91	0.77
Cyclothymic temperament	0.81	0.85
Hyperthymic temperament	0.93	0.80
Irritable temperament	0.76	0.82
Anxious temperament	0.73	0.84

The investigation was conducted in conformity with the Turkish laws about questionnaire research.

2.3. Subjects

The sample of the study consisted of 658 clinically-well Turkish subjects of both sexes, 43% males ($n=286$) and 57% females ($n=372$). Mean age of all subjects was 31.6 (S.D.=11.6), and its range was between 18 and 88 years. Fifty-two percent of the sample was single, 41% married, and 3% was separated or divorced. The majority (65%) had a university degree, 26% had completed high school, and 8% had elementary school diplomas. Our sample consisted of subjects in the university circle in İzmir: university students of various orientations, workers, professionals, academic faculty, artists, retired or unemployed people and housewives. The measure was administered to the volunteering subjects by two of the authors (S.K.,

Table 2
Psychometrically weak items that were deleted

Depressive temperament
Item #16. I am a hard working person
Item #18. It is natural for me to be neat and organized
Cyclothymic temperament
Item #37. My need for sleep varies a lot from just a few hours to more than 9 h
Item #41. I often have a strong urge to do outrageous things
Hyperthymic temperament
Item #63. Normally less than 6 h of sleep is enough for me
Irritable temperament
Item #75. I like to tease people, even those I hardly know
Item #82. I could be a revolutionary
Item #84. (Women only): I have attacks of uncontrollable rage right before my periods
Anxious temperament
Item #103. I am, by nature, a very cautious person
Item #104. I often wake up at night afraid that burglars are in the house

M.A.), after making sure on the basis of a brief clinical interview that they were free of major mental disorders.

2.4. Statistical analyses

- Test–retest reliability was conducted to find out if the test yields the same scores when administered in various conditions. The temperaments of the sub-

Table 3
Factor loadings of each temperament as (originally conceived) and suggested interpretations

Factor	Items	Depressive temperament
1st	9, 10 and 11	Introversion
2nd	5, 6, 8 and 17	Low self-esteem
3rd	1, 2, 3 and 4	Pessimistic
4th	7, 12, 13 and 19	Sensitivity
5th	14 and 15	Altruism
Factor	Items	Cyclothymic temperament
1st	22, 23, 24, 25, 29 and 30	Variability
2nd	31, 32, 33, 34, 35 and 38	Cyclicality
3rd	26, 28 and 42	Instability
4th	27, 36, 39 and 40	Intensity
Factor	Items	Hyperthymic temperament
1st	43, 44, 45, 46 and 55	Fun loving
2nd	47, 48, 49, 50, 51 and 58	High self-esteem
3rd	59, 60, 61, 62	Narcissistic
4th	52, 53	Risk-taking/ adventuresome
5th	54, 56 and 57	Sociable
Factor	Items	Irritable temperament
1st	68, 69, 70 and 71	Restlessness
2nd	72, 73, 74 and 77	Aggressivity
3rd	67 and 76	Critical of others
4th	78, 79, 80 and 83	Dysphoric emotionality
5th	64, 65, 66 and 81	Complaining
Factor	Items	Anxious temperament
1st	85, 86, 87, 88 and 89	Worrying about kin
2nd	91, 92, 101 and 102	Inability to relax
3rd	90, 93, 105 and 106	Somatic anxiety
4th	98, 99, 100, 107, 108, 109 and 110	Fear-prone
5th	94, 95, 96 and 97	Autonomic anxiety

jects are not supposed to change, so the scores at two times should be correlated. The test was administered to the same population twice in 2 weeks intervals for a subset of 30 subjects.

- The alpha value of each item on the scale was evaluated and correlated with Cronbach-Alpha. The results of this analysis would serve to indicate that items of each scale represent related traits necessary for defining the five postulated constructs.
- Factor analysis with varimax PCA rotation was carried out to assess the internal consistencies of each of the subscales.
- We evaluated socio-demographic characteristics. Preliminary cut-off points were determined for each subscale via calculation of *z*-scores corresponding to +2S.D. Subjects who had higher scores than cut-off were considered as having the specified dominant temperaments.

3. Results

3.1. Subjects

The measure was administered to 1000 subjects. Thirty four percent of the administered questionnaires were either incomplete or not returned. The analyses reported herein pertain to the 658 (66%) protocols which were complete. Completers and not-completers were demographically comparable.

3.2. Reliability

As shown in Table 1, test–retest reliability ranged from 0.73 to 0.93 and Cronbach-Alpha coefficients from 0.75 to 0.84.

Item analysis indicated that 10 items had alpha values of <0.2 (Table 2). The homogeneity of the scale will improve if these were removed from the scale. Some are sleep related items, others appear to be social desirability items. These items were deleted from further consideration in the final 100 item questionnaire TEMPS-A, Turkish version (see Appendix A, where these items are marked by an asterisk, *).

3.3. Validity

The construct validity of the scale was studied using exploratory factor analysis (PCA type 2, varimax rotation). Factor analysis of all items yielded five main factors (Appendix A). As can be seen, factor loadings of depressive and anxious (cognitive) items overlap significantly and form a “cluster.” However, another smaller factor loading with certain somatic items of the anxious temperament stands out as a separate cluster, without any overlap with other temperaments, which we decided to label the “nervous temperament”. Cyclothymic, irritable and hyperthymic temperaments formed largely independent factors (Table 3).

In order to obtain additional support for the validity of the component traits of each tempera-

Table 4
Means and standard deviations of scores; frequency distributions; cut-off points of *z*-scores

		– 3S.D.	– 2S.D.	– 1S.D.	+ 1S.D.	+ 2S.D.	+ 3S.D.	+ 4S.D.
Depressive temperament	<i>n</i>	0	150	232	201	54	17	4
	(%)	–	(22.8)	(35.2)	(30.6)	(8.2)	(2.7)	(0.4)
	Score	–	–	–	7–9	10–12	13–15	16–18
Cyclothymic temperament	<i>n</i>	0	152	228	157	80	11	0
	(%)	–	(23.1)	(34.6)	(23.7)	(13.9)	(1.7)	–
	Score	–	–	–	9–13	14–17	18–19	–
Hyperthymic temperament	<i>n</i>	20	95	260	189	86	8	0
	(%)	(3.1)	(14.4)	(39.6)	(28.7)	(13)	(1.2)	–
	Score	–	–	–	12–15	16–19	20	–
Irritable temperament	<i>n</i>	0	160	235	163	76	23	1
	(%)	–	(24.3)	(35.6)	(24.8)	(11.6)	(3.5)	(0.2)
	Score	–	–	–	5–8	9–12	13–16	17–18
Nervous temperament	<i>n</i>	0	152	256	158	69	20	3
	(%)	–	(23)	(38.8)	(23.9)	(10.6)	(3.1)	(0.6)
	Score	–	–	–	8–12	13–17	18–21	22–24

Table 5
The comparisons of temperament scores according to gender

	Male	Female	<i>P</i> *
Depressive temperament	13.9 (3.5)	14.8 (4.1)	0.073
Cyclothymic temperament	18.8 (2.8)	18.7 (3.2)	0.643
Hyperthymic temperament	19.7 (1.6)	17.6 (2.1)	0.012
Irritable temperament	21.4 (2.6)	20.6 (2.8)	0.500
Nervous–anxious temperament	20.6 (3.1)	22.3 (3.6)	0.025

* Mann–Whitney *U*.

ment (Akiskal and Mallaya, 1987; Akiskal et al., 1978), factor analysis for each temperament was also performed. The items of each temperament loaded on five or six subfactors (Table 3). Suggested interpretation of each of these subfactors are provided. The factors obtained for each temperament are essentially compatible with the original Akiskal and Mallaya (1987; Akiskal and Akiskal (1992) criteria for each of the “classic,” as well as to the subsequently added generalized anxious (Akiskal, 1998) temperaments.

3.4. Determining cut-off scores

Dominant temperament was determined according to their *z*-scores: S.D. +1, mildly high; S.D. +2, moderately high; S.D. +3, extremely high. The cut-off scores were evaluated according to the *z*-scores above the positive 2S.D. (Table 4).

3.5. Gender and age effects on prevalence

The relationship between temperament and gender was assessed with respect to the scores obtained for each temperament and the dominant temperament of the subject as well (Table 5). Males had significantly higher scores than females for the

Table 6
Rates of dominant temperaments by gender

	Male		Female		Total	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Depressive temperament	9	1.5	12	2.7	21	3.1
Cyclothymic temperament	4	0.7	7	1.2	11	1.7
Hyperthymic temperament	4	0.7	4	0.7	8	1.2
Irritable temperament	14	2.4	10	2	24	3.7
Anxious temperament	8	1.4	15	2.7	23	3.7
Total	39	6.7	48	9.1	87	13.4

Table 7
Rates of dominant temperaments by age groups*

	<30 years		31–40 years		>41 years	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Depressive temperament	12	2.1	5	0.8	4	0.7
Cyclothymic temperament	6	1	3	0.5	2	0.3
Hyperthymic temperament	5	0.8	2	0.3	1	0.1
Irritable temperament	14	2.4	7	1.2	3	0.5
Anxious temperament	14	2.4	6	1	3	0.5

* All comparisons non-significant.

hyperthymic temperament ($U=200.21$, $P=0.012$), whereas nervous–anxious temperament scores were significantly higher in women ($U=158.32$, $P=0.025$). There was no statistically significant gender difference in the frequency of dominant depressive, cyclothymic, and irritable temperaments (Table 6). Prevalence of the dominant temperaments combining the two sexes are as follows: 3.1% for depressive temperament, 1.7% for cyclothymic temperament, 1.2% for hyperthymic temperament, 3.7% for irritable temperament, 3.7% for the anxious temperament.

No significant difference was found among three age groups (age under 30, between 31 and 40 and over 41) in mean temperament scores. However, as the age increased, the temperament scores tended to decrease (Table 7).

4. Discussion

The findings of this study indicate that TEMPS-A (Turkish Version) is a reliable instrument that is suitable for future studies in Turkey (Vahip et al., *in press*).

Overall, the hypothesized 5-factor structure of the TEMPS was supported, each scale contributing largely to its own factor. However, factor analysis findings indicate that depressive and anxious (cognitive) traits overlap considerably. Although this finding implies that depressive and anxious traits coexist in the community, a dominant depressive temperament—including trait cognitive anxiety—could be delineated; we also found evidence for a distinct “nervous”–anxious factor. There was also a minor overlap of the dominant hyperthymic with several irritable traits. Dominant cyclothymic and

irritable temperaments had minimal overlap with items from other subscales.

TEMPS-A (Turkish Version), like its original English counterpart, has good internal consistency. This was enhanced by removing items with low factorial loading from the instrument. We found that several sleep items were not useful as trait measures. Curiously, being a “short sleeper” did not characterize the hyperthymic. Excessive need for sleep characterized not the depressive, but the cyclothymic, where it may tap the tendency to fatigue and disposition to retarded depression. As expected, unrestful sleep and initial insomnia characterized the anxious temperament. The lack of traitness for some of the sleep items has been found previously in the Italian sample as well (Placidi, 1998). Future studies are needed to clarify this question. Another item with weak performance was, “I am a hardworking person” (originally in the depressive scale). This item has a positive social valence in the Turkish culture, and does not appear to tap depressive traits. Other items could also be influenced by cultural factors. Thus the self-concept of being neat and organized (#18, originally in the depressive temperament) appears like a “social desirability” item only endorsed by a positive temperament like the hyperthymic, and we have elected to delete it as well. It is expected that the urge to engage in outrageous acts to be characteristic of the cyclothymic (#41), yet it was endorsed by the hyperthymic; as other items (such as narcissistic traits) seemed to measure the same idea, we also elected to delete it. On the other hand, the theoretical structure of the TEMPS-A which mixes adaptive social traits and affect-laden items is supported by the subfactorial loading for each temperament (Table 4). These include, among others, fun loving, narcissism, sociability and worrying about kin.

Now brief comments about each of the constituent temperaments. The depressive temperament has been validated and standardized on the basis of this first psychometric study in Turkey, yielding a prevalence of 3.1%. In previous studies based in clinical populations (Klein, 1990; Phillips et al., 1998) using similar criteria in other populations, much higher rates of prevalence have been reported, possibly due to the use of different cut-offs. Rates similar to the present one

were obtained in the study with TEMPS-I Scale in an Italian student population (von Zerssen and Akiskal, 1998; Placidi, 1998).

The prevalence for cyclothymia in a student population was reported as 4–6% (Depue et al., 1981). In the study that was done in Italy to determine the reliability and validity of the TEMPS-I, the prevalence of this temperament was 6.4% (Placidi, 1998). We found that the prevalence of cyclothymic temperament in our population was 1.7%, and had a good validity. The difference can be due to interview versus self-rated measures, or in part explained by the much younger Italian sample. Alternatively, there can be real population differences.

In the Turkish sample, the prevalence of dominant hyperthymic temperament was 1.2%, which was validated and standardized. These data cohere with two other population studies. Thus, Eckblad and Chapman (1986) found that intermittent hypomanic characteristics occurred in 1% in 1519 college students. Wicki and Angst (1991) found a prevalence of 4% for hypomania, of which one out of four (1% of the population) was chronically hypomanic, i.e. “hyperthymic”. However, in the study that was carried out in Italy, none of the subjects obtained scores in the +2S.D. range; it appeared as though many Italian students were hyperthymic (between the 1st and 2nd + S.D.), and for that reason standardization could not be carried out in that population. Further studies are needed to examine the relative merits of interview and self-rated versions in assessing the hyperthymic temperament.

The distribution of irritable temperament was non-Gaussian and unstandardized in the Italian TEMPS-I study (Placidi, 1998), it was interestingly well standardized and validated in the Turkish sample, accounting for 3.7% of subjects. Such a difference could stem from better operationalization of this temperament in TEMPS-A compared with TEMPS-I, though true population differences cannot be ruled out.

The prevalence of the dominant “nervous” temperament in the Turkish sample was 3.7%. This rate is compatible with epidemiologic data on generalized anxiety (Wittchen et al., 1994).

Depressive, irritable and “nervous” temperaments were the most common temperaments in the Turkish

sample, and cyclothymic and hyperthymic, the least common. As already pointed out, except for concordance with the depressive, these rates are in some ways in sharp contrast with those from Italy. The present findings are an important beginning, since this is the first study in Turkey that assessed affective temperaments. Therefore, replication of this study is needed on a larger and more representative national sample to decide whether such differences are due to population/cultural differences, or emanate from methodologic factors.

As for the distribution by gender, in our population men scored significantly higher on hyperthymic temperament, and this finding is similar with other studies in clinical populations (Akiskal, 1984; von Zerssen and Akiskal, 1998; Perugi, 1990). As expected, women scored significantly higher on the “nervous” temperament and also had more dominant anxious temperament traits than males.

A significant difference with regard to age groups was not obtained, however it was observed that extreme scores in older ages were rare. Although this finding was not statistically significant, it implied that as age increases, the temperament scores might decrease. Even this trend might be a measurement

artifact, given the relatively small subsample in the elderly in the present study.

5. Conclusions and future directions

We tentatively conclude that anxious, irritable and depressive temperaments are the most dominant temperaments in Turkey: The “nervous” temperament is more frequent in females, irritable temperament in males.

This is a first attempt to psychometrically evaluate the TEMPS-A (Turkish Version). It is a reliable and valid instrument in our hands. Since it is a self-report instrument, it is convenient to administer it in large epidemiological studies. In clinical and genetic studies, however, reliability and validity of the instrument are of greater importance, therefore, further development and standardization of semi-structured format of the 100-item TEMPS-A could improve the quality of future studies in clinical populations from different socio-economic classes with different literacy rates—such as that in Turkey—where investigators often need to read the questions to the subjects.

Appendix A. Factor structure after varimax (normalized) rotation of Turkish TEMPS-A

The original English text TEMPS-A is given in italics to facilitate understanding.

		Factors				
		1	2	3	4	5
1 D	Üzgün, mutsuz bir insanım. <i>I am a sad, unhappy person</i>	0.51	0.14	−0.12	0.13	−0.20
2 D	İnsanlar bana olayların komik yanlarını göremediğimi söylerler. <i>People tell me I am unable to see the lighter side of things</i>	0.41	0.13	−	−	−0.12
3 D	Hayatım boyunca çok acıktım. <i>I have suffered a lot in life</i>	0.41	0.15	−	−	−
4 D	İşlerin sonu sıklıkla kötüye varır diye düşünürüm. <i>I think things often turn out for the worst</i>	0.54	0.25	−	0.17	−
5 D	Kolay pes ederim. <i>I give up easily</i>	0.27	0.23	−	−0.31	−
6 D	Kendimi bildim bileli başarısız hissetmişimdir. <i>As long as I can remember, I have felt like a failure</i>	0.32	0.18	−0.29	0.15	−
7 D	Başkalarının sorun saymadıkları konularda ben kendimi daima sudanmışım. <i>I have always blamed myself for what others might consider no big deal</i>	0.53	0.21	−0.23	0.10	0.11

(continued on next page)

Appendix A (continued)

8 D	Başka insanlar kadar çok enerjim olduğunu sanmıyorum. <i>I do not seem to have as much energy as other people</i>	0.31	0.22	– 0.37	–	0.12
9 D	Değişikliği pek sevmeyen tipte bir kişiyim. <i>I am the kind of person who does not like change very much</i>	0.25	–	– 0.19	–	–
10 D	Grup içinde konuşmaktansa, başkalarını dinlemeyi tercih ederim. <i>In a group, I would rather hear others talk</i>	0.27	–	– 0.31	– 0.17	0.13
11 D	Sıklıkla önceliği başkalarına veririm. <i>I often give into others</i>	0.25	0.16	– 0.26	– 0.24	0.13
12 D	Yeni insanlarla karşılaştığımda kendimi oldukça rahatsız hissedirim. <i>I feel very uneasy meeting new people</i>	0.38	0.24	– 0.21	–	–
13 A	Eleştirildiğimde ya da reddedildiğimde duygularım çok kolay zedelenir. <i>My feelings are easily hurt by criticism or rejection</i>	0.19	0.12	– 0.18	–	0.36
14 D	Ben her zaman güvenilebilecek tipte bir insanım. <i>I am the kind of person you can always depend on</i>	–	–	– 0.19	0.22	0.14
15 A	Başkalarının ihtiyadlarını kendiminkilerin önüne koyarım. <i>I put the needs of others above my own</i>	0.16	0.23	– 0.18	–	0.29
16 *	Çok çalışan bir kişiyim. <i>I am a hard working person</i>	–	–	– 0.19	0.12	0.14
17 A	İşlerin başında olmaktansa başkasının altında çalışmayı tercih ederim. <i>I would rather work for someone else than be the boss</i>	–	–	– 0.36	– 0.11	0.23
18 *	Planlı-programlı ve düzenli olmak benim yapımda vardır. <i>It is natural for me to be neat and organized</i>	0.14	– 0.11	0.51	– 0.21	0.12
19 D	Hıfır şeyden emin olamayan tipte bir kişiyim. <i>I am the kind of person who doubts everything</i>	0.37	0.19	– 0.22	0.10	0.28
20 D	Cinsel arzularım her zaman az olmuştur. <i>My sex drive has always been low</i>	0.11	–	– 0.31	–	0.11
21 C	Normal olarak günde 9 saatten fazla uykuya ihtiyacım vardır. <i>I normally need more than 9 h of sleep</i>	0.15	0.33	–	–	–
22 C	Çoğu zaman sebepsiz yorgunluk hissedirim. <i>I often feel tired for no reason</i>	0.21	0.49	–	–	0.15
23 C	Ruh halimde ve enerjimde ani değişiklikler olur. <i>I get sudden shifts in mood and energy</i>	–	0.59	–	0.19	0.10
24 C	Ruh halim ve enerjim ya yukarılarda ya da aşağılardadır, ender olarak ikisinin arasındadır. <i>My moods and energy are either high or low, rarely in between</i>	0.22	0.58	0.15	–	–
25 C	Ortada görünür bir neden yokken zihnim bazen çok açık bazen de donuk olur. <i>My ability to think varies greatly from sharp to dull for no apparent reason</i>	–	0.64	– 0.12	–	0.18
26 C	Birisini gerçekten çok sevebilir ve sonra ona olan ilgimi tamamen kaybedebilirim. <i>I can really like someone a lot, and then completely lose interest in them</i>	–	0.44	0.16	0.18	– 0.20
27 I	Sıklıkla insanlara parlarım ve sonra suduluk duyurum. <i>I often blow up at people and then feel guilty about it</i>	–	0.37	0.14	0.38	–

Appendix A (continued)

28 C	Sıklıkla birşeylere başlar ve onları bitirmeden ilgimi kaybederim. <i>I often start things and then lose interest before finishing them</i>	–	0.47	–	0.19	–
29 C	Ruh halim (duygularım) sebepsiz yere sık sık değişir. <i>My mood often changes for no reason</i>	0.19	0.63	–	0.25	–
30 C	Canlılık ve uyuşukluk arasında sürekli gider gelirim. <i>I constantly switch between being lively and sluggish</i>	0.21	0.62	–	0.16	–
31 C	Bazen yatağa öökkün girer, fakat sabah müthiş yi hissederek uyanırım. <i>I sometimes go to bed feeling down, but wake up in the morning feeling terrific</i>	–	0.41	–	0.26	–
32 C	Bazen yatağa kendimi çok iyi hissederek girer ve sabahleyin yaşamın yaşanmaya değer olmadığı duygusuyla uyanırım. <i>I sometimes go to bed feeling great, and wake up in the morning feeling life is not worth living</i>	0.26	0.52	–	–	–
33 C	Bana olaylar karşısında sıklıkla karamsarlığa kapıldığım ve eski mutlu zamanları unuttuğum söylenir. <i>I am told that I often get pessimistic about things, and forget previous happy times</i>	0.32	0.42	–0.10	0.10	–
34 C	Aşırı kendine güven ile bir türlü kendinden emin olamama duyguları arasında gider gelirim. <i>I go back and forth between feeling overconfident and feeling unsure of myself</i>	0.21	0.55	–0.14	–	–
35 C	Dışa dönüklükle içe kapanma arasında gider gelirim. <i>I go back and forth between being outgoing and being withdrawn from others</i>	0.22	0.50	–0.16	0.10	–
36 C	Tüm duyguları yoğun olarak yaşarım. <i>I feel all emotions intensely</i>	–	0.27	0.16	0.11	–
37 *	Birkac saat ile 9 saat arasında değişen sürelerde uykuya ihtiyac duyarım. <i>My need for sleep varies a lot from just a few hours to more than 9 h</i>	–	0.10	–	–	–
38 C	Bazen herşeyi çok canlı, bazen yaşamdan yoksun denecek kadar renksiz algıları. <i>The way I see things is sometimes vivid, but at other times lifeless</i>	0.25	0.56	–	–	–
39 C	Aynı anda hem mutsuz, hem de mutlu olabilen tipte bir kişiyim. <i>I am the kind of person who can be sad and happy at the same time</i>	0.15	0.44	0.18	–	–
40 C	Başkalarının ulaşılmaz saydıkları şeyler hakkında uzun uzun hayaller kurarım. <i>I daydream a great deal about things that other people consider impossible to achieve</i>	–	0.34	0.22	0.15	0.10
41 *	Sıklıkla kuraldışı, topluma aykırı şeyler yapmak için güdü bir istek duyarım. <i>I often have a strong urge to do outrageous things</i>	0.10	0.20	0.42	–	–0.18
42 C	Kolay asık olup, kolay vazgeçen bir kişiyim. <i>I am the kind of person who falls in and out of love easily</i>	–	0.39	0.23	–	–
43 H	Çoğunlukla havamda ya da neşeli bir ruh halindeyimdir. <i>I am usually in an upbeat or cheerful mood</i>	–0.37	–	0.41	–	0.18

(continued on next page)

Appendix A (continued)

44 H	Yaşam, sonuna kadar tadını ıkardığım bir şölendir. <i>Life is a feast which I enjoy to the fullest</i>	– 0.23	– 0.13	0.47	–	0.13
45 H	Fikra anlatmayı, espriler yapmayı severim, insanlar bana şakacı olduğumu söylerler. <i>I like telling jokes, people tell me I am humorous</i>	– 0.22	–	0.39	–	0.12
46 H	Herşeyin zamanla iyi olacağına inanan tipte bir kişiyim. <i>I am the kind of person who believes everything will eventually turn out all right</i>	– 0.17	–	0.32	– 0.28	0.20
47 H	Kendime müthiş güvenirim. <i>I have great confidence in myself</i>	–	–	0.68	–	–
48 H	Sıklıkla büyük fikirler üretirim. <i>I often get many great ideas</i>	–	–	0.65	–	–
49 H	Her zaman birşeylerle meşgulümdür. <i>I am always on the go</i>	–	–	0.56	–	– 0.12
50 H	Birçok işi hem de yorulmadan yapabilirim. <i>I can accomplish many tasks without even getting tired</i>	–	–	0.56	–	– 0.12
51 H	Konuşmaya doğuştan yetenekliyim. Konuşmam başkaları için ikna edici, etkileyici ve ilham vericidir. <i>I have a gift for speech, convincing and inspiring to others</i>	–	–	0.55	–	–
52 H	Riskli bile olsalar yeni projelere atlamaya bayılırım. <i>I love to tackle new projects, even if risky</i>	–	0.16	0.51	–	– 0.21
53 H	Bir şey yapmayı bir kez kafama koyduğumda, beni hiçbir şey durduramaz. <i>Once I decide to accomplish something, nothing can stop me</i>	–	–	0.55	–	–
54 H	Doğru dürüst tanımadığım insanlarla bile son derece rahatım. <i>I am totally comfortable even with people I hardly know</i>	– 0.12	–	0.50	–	–
55 H	İnsanlarla birlikte olmayı çok severim. <i>I love to be with a lot of people</i>	– 0.19	0.10	0.32	–	0.20
56 H	İnsanlar bana sıklıkla burnumu başkalarının işine soktuğumu söylerler. <i>People tell me that I often get my nose into others business</i>	–	–	0.37	–	0.10
57 H	Cömertim ve başkaları için bol para harcarım. <i>I am generous, and spend a lot of money on other people</i>	–	0.18	0.29	–	–
58 H	Birçok alanda yetenekli ve uzmanım. <i>I have abilities and expertise in many areas</i>	–	–	0.51	–	–
59 H	Canımın istediğini yapma hakkım ve ayrıcalığım olduğunu hissederim. <i>I feel I have the right and privilege to do as I please</i>	–	0.22	0.43	0.19	–
60 H	İşin patronu, “tepedeki adam” olmayı seven tipte bir kişiyim. <i>I am the kind of person who likes to be the boss</i>	–	–	0.36	0.25	–
61 H	Birisiyle bir konu üzerinde anlaşmadığım zaman ateşli bir tartışmaya girebilirim. <i>When I disagree with someone, I can get into a heated argument</i>	–	–	0.45	0.18	–
62 H	Cinsel isteklerim daima fazladır. <i>My sex drive is always high</i>	–	–	0.50	0.11	– 0.17

Appendix A (continued)

63*	Normalde 6 saatten az uyku bana yeter. <i>Normally I can get by with less than 6 h of sleep</i>	0.10	–	–	–	– 0.15
64 I	Huysuz (sinirli) bir kişiyim. <i>I am a grouchy (irritable) person</i>	0.19	–	–	0.63	–
65 I	Bir türlü hoşnut olmayan tabiatla bir kişiyim. <i>I am by nature a dissatisfied person</i>	0.28	–	–	0.52	–
66 I	Çok yakınıyorum. <i>I complain a lot</i>	0.31	–	– 0.14	0.49	0.12
67 I	Başkalarını çok eleştiririm. <i>I am highly critical of others</i>	–	–	–	0.52	0.13
68 I	Kendimi sıklıkla patlamaya hazır, gergin hissedirim. <i>I often feel on edge</i>	0.47	0.20	–	0.54	–
69 I	Kendimi sıklıkla keman yayı gibi gerilmiş hissedirim. <i>I often feel wound up</i>	0.38	0.12	–	0.48	–
70 I	Adeta hayatımı anlayamadığım, hoş olmayan bir huzursuzluk yönlendiriyor. <i>I am driven by an unpleasant restlessness that I do not understand</i>	0.28	0.27	– 0.14	0.53	–
71 I	Sık sık öylesine sinirleniyorum ki, gözüm hic bir şey görmüyor. <i>I often get so mad that I will just trash everything</i>	0.23	0.23	–	0.51	–
72 I	Terslendiğimde kavga edebilirim. <i>When crossed, I could get into a fight</i>	–	0.10	–	0.48	0.12
73 I	İnsanlar bana hic yokken parladığımı söylerler. <i>People tell me I blow up out of nowhere</i>	0.15	0.23	0.14	0.50	–
74 I	Sinirlendiğimde insanlara bağırırım. <i>When angry, I snap at people</i>	–	0.17	–	0.47	–
75 *	Az tanısam bile insanlara takılmayı, şakalaşmayı severim. <i>I like to tease people, even those I hardly know</i>	– 0.17	0.14	0.18	0.16	–
76 I	İğneleyici şakalarım beni zor durumda bırakır. <i>My biting humor has gotten me into trouble.</i>	–	0.14	0.18	0.42	–
77 I	O kadar öfkelenebilirim ki birilerine zarar verebilirim. <i>I can get so furious I could hurt someone</i>	0.10	0.21	–	0.30	– 0.12
78 I	Eşimi (ya da sevgilimi) o kadar kıskanırım ki buna dayanamıyorum. <i>I am so jealous of my spouse (or lover), that I cannot stand it</i>	0.14	0.20	0.18	0.35	0.21
79 I	Küfürbaz olarak bilinirim. <i>I am known to swear a lot</i>	– 0.10	–	–	0.31	–
80 I	Birkac kadeh idkiyle saldırganlaştığım söylenmiştir. <i>I have been told that I become violent with just a few drinks</i>	–	0.14	0.10	0.19	–
81 I	Çok kuşkucu bir kişiyim. <i>I am a very skeptical person</i>	0.28	0.15	–	0.40	–
82 *	Devrimci bir ruha sahibim. <i>I could be a revolutionary</i>	–	–	–	0.11	–
83 I	Cinsel isteklerim sıklıkla o kadar yoğundur ki gerçekten rahatsızlık yaratır. <i>My sex drive is often so intense that it is truly unpleasant</i>	–	–	0.21	0.32	–
84 *	(Yalnızca bayanlar): Adet dönemlerimin hemen öncesinde kontrol edemediğim öfkelerim olur. <i>(Women only): I have attacks of uncontrollable rage right before my periods</i>	0.10	–	–	0.10	–
85 D	Kendimi bildim bileli endişeli biriyim. <i>I have been a worrier for as long as I can remember</i>	0.60	–	–	–	0.34

(continued on next page)

Appendix A (continued)

86 D	Her zaman endişelenecek bir şey bulurum. <i>I am always worrying about one thing or another</i>	0.65	–	–	0.13	0.22
87 D	Başkalarının ufak tefek saydığı günlük şeyler hakkında endişelenir durumum. <i>I keep on worrying about daily matters that others consider minor</i>	0.64	–	–	–	0.25
88 D	Endişelenmemin önüne geçemiyorum. <i>I cannot help worrying</i>	0.69	–	–0.17	0.12	0.39
89 D	Birçok insan bana bu kadar endişelenmememi söylemiştir. <i>Many people have told me not to worry so much</i>	0.71	–	–	–	0.29
90 A	Zorda kaldığımda oğu kez kafam durur, bloke olurum. <i>When stressed, my mind often goes blank</i>	0.21	0.11	–0.16	–	0.43
91 D	Gevşemeyi beceremiyorum. <i>I am unable to relax</i>	0.45	0.18	–	0.10	0.13
92 D	Sık sık içimde huzursuz bir kıpırtı hissedirim. <i>I often feel jittery inside</i>	0.53	0.23	–	0.22	0.13
93 A	Zor durumda kaldığımda, sıkıldığımda sıklıkla ellerim titrer. <i>When stressed, my hands often tremble</i>	0.21	0.20	0.10	–	0.33
94 A	Sık sık midem bozulur. <i>I often have an upset stomach</i>	0.12	–	–	0.10	0.49
95 A	Heyecanlandığımda ishal olabilirim. <i>When I am nervous, I may have diarrhea</i>	–	–	–	–	0.53
96 A	Heyecanlandığımda sıklıkla bulantı hissedirim. <i>When I am nervous, I often feel nauseous</i>	–	0.14	–	–	0.46
97 A	Heyecanlandığımda tuvalete daha sık gitmek zorunda kalırım. <i>When I am nervous, I have to go to the bathroom more often</i>	–	–	–0.13	0.11	0.53
98 D	Birisi eve geç kaldığı zaman basına bir kaza gelmiş olabileceğinden korkarım. <i>When someone is late coming home, I fear they may have had an accident</i>	0.36	0.11	–	–0.21	0.12
99 D	Sıklıkla ailemden birileri ciddi bir hastalığa yakalanacak diye çok korkarım. <i>I am often fearful of someone in my family coming down with a serious disease</i>	0.43	0.32	0.11	–0.21	0.22
100 D	Devamlı olarak aile üyelerinden biriyle ilgili kötü bir haber alacakmışım gibi geliyor. <i>I am always thinking someone might break bad news to me about a family member</i>	0.49	0.25	0.12	–0.24	0.21
101 A	Uykum dinlendirici değil. <i>My sleep is not restful</i>	0.36	0.23	–	–	0.21
102 A	Sıklıkla uykuya dalmakta güdük ækerim. <i>I frequently have difficulty falling asleep</i>	0.32	0.22	–	–	0.12
103 *	Ben tabiat olarak çok temkinli bir kişiyim. <i>I am, by nature, a very cautious person</i>	0.11	0.12	–	0.12	–
104 *	Sıklıkla geceleri eve hırsız girmiş diye korkarak uyanırım. <i>I often wake up at night afraid that burglars are in the house</i>	0.16	0.16	–	–	–
105 A	Zor durumda kaldığımda, sıkıldığımda hemencecik basım ağrır. <i>I easily get headaches when stressed</i>	–	0.11	–	–	0.56
106 A	Zor durumda kaldığımda, sıkıldığımda göğsüm sıkışır. <i>When stressed, I get an uncomfortable feeling in my chest</i>	0.13	–	–	–	0.57
107 D	Kendimi güvende hissetmiyorum. <i>I am an insecure person</i>	0.45	0.23	–	0.18	–
108 D	Günlük işleyişteki küçük değişiklikler bile beni çok zorlar. <i>Even minor changes in routine stress me highly</i>	0.31	0.18	–	0.25	0.22

Appendix A (continued)

109 D	Araba kullanırken yanlış herhangi bir şey yapmasam bile, polis beni durduracakmış gibi bir korku duyarım. <i>While driving, even when I have not done anything wrong I fear that police may stop me</i>	0.40	0.17	–	–	0.11
110 A	Ani sesler beni kolayca irkiltir. <i>Sudden noises startle me easily</i>	0.19	0.24	–0.15	–	0.38

D, item was selected for the depressive temperament subscale; C, item was selected for the cyclothymic temperament subscale; H, item was selected for the hyperthymic temperament subscale; I, item was selected for the irritable temperament subscale; A, item was selected for the anxious temperament subscale; *, item was discarded from TEMPS-A TV.

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